SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	G Rahul	**Reg. No.:**	192572008
Date:	22-10-2026	**Duration:**	60 minutes

Code of Conduct

*I **G Rahul (192572008)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **G Rahul**

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario** o Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements** o List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

3. **Identify Actors and Use Cases** o Identify all primary and secondary actors interacting with the system. o List the major use cases associated with each actor.
   - o Draw a Use Case Diagram representing the interactions between actors and the system.

4. **Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:
   - o Introduction o Problem Description o Scope of the System o Functional Requirements o Non-Functional Requirements o Use Case Descriptions o Data Requirements o External Interface Requirements o System Features

5. **Identify Assumptions and Constraints** o List the assumptions considered while developing the system.
   - o Identify technical, operational, economic, legal, and time-related constraints affecting the project.

6. **Validate Requirement Completeness and Consistency** o Verify whether all stakeholder requirements have been addressed. o Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - o Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and wellclassified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.

Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is wellstructured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Document Control

Item	Description
Project Title	Smart Classroom Management System
Document	Scenario-Based Requirement Analysis and Software Requirement Specification
Version	1.0
Prepared for	CSA1009– Software Engineering
Purpose	To define the system scope, stakeholders, requirements, use cases, interfaces, constraints, and validation checks.

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Note: In Microsoft Word, the table of contents can be updated or converted to an automatic TOC if required by the instructor.

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) describes the requirements for a Smart Classroom Management System. It provides a common understanding for students, faculty members, administrators, and developers about what the system must do and the quality standards it should meet. The document is intended to guide design, implementation, testing, and future maintenance.

1.2 Intended Audience

- Project evaluators and faculty members.
- System administrators and academic coordinators.
- Faculty members who manage courses, schedules, materials, and assignments.
- Students who access learning resources, submit work, and participate in classroom activities.
- Developers and testers responsible for building and validating the application.

1.3 Business Domain

The system belongs to the education and academic administration domain. It supports routine classroom and course activities through a centralized digital platform, reducing dependence on scattered notices, paper records, and separate communication channels.

1.4 Definitions and Abbreviations

Term	Meaning
SRS	Software Requirement Specification
Admin	Authorized person who manages users, courses, and system settings
Faculty	Teaching staff member who manages course activities
Student	Learner enrolled in one or more courses
Use case	A description of an interaction that helps an actor achieve a goal
FR / NFR	Functional Requirement / Non-Functional Requirement

2. Scenario Study and Problem Analysis

2.1 Scenario Description

An educational institution manages classroom schedules, course materials, assignments, announcements, student participation, and faculty activities. When these activities are handled through separate spreadsheets, messaging groups, printed notices, or manual registers, information can become inconsistent or difficult to locate. Students may miss schedule changes or deadlines, while faculty may spend extra time distributing materials, collecting assignments, and maintaining participation records. The proposed Smart Classroom Management System provides one role-based platform to coordinate these activities and make current information available to authorized users.

2.2 Existing Challenges

- Classroom schedules and updates may be distributed through multiple channels.
- Students may have difficulty finding the latest course materials and assignment instructions.
- Assignment submission and deadline tracking may require manual follow-up.
- Announcements may not reach every intended student in time.
- Participation and faculty activity records may be stored in separate registers or files.
- Duplicate data entry and manual reporting can increase workload and errors.
- There may be limited visibility into pending assignments, upcoming classes, and recent announcements.

2.3 Stakeholders

Stakeholder	Interest / Responsibility
Students	View schedules and materials, read announcements, submit assignments, and review feedback.
Faculty members	Manage course content, schedules, assignments, announcements, participation, and relevant activity records.
Administrator	Create and manage accounts, courses, permissions, and system-wide settings.
Academic coordinator / department staff	Review schedule coordination and academic activity summaries where permission is granted.
Technical support / system maintainer	Maintain availability, troubleshoot issues, back up data, and apply updates.
Institution management	Use authorized summaries to understand adoption and

	operationa

2.4 System Objectives

- Centralize classroom and course-related information.
- Make schedules, learning resources, assignments, and announcements accessible to authorized users.
- Support assignment creation, submission, deadline tracking, and feedback.
- Provide a consistent way to record and review student participation.
- Help faculty organize their academic activities and reduce repetitive administrative work.
- Protect student and faculty information through authentication and role-based access.
- Improve the accuracy, traceability, and timeliness of classroom communication.

3. Scope of the System

3.1 In Scope

- Secure login and role-based access for administrators, faculty, and students.
- User, course, and enrollment management.
- Classroom timetable creation, viewing, and change notifications.
- Uploading, organizing, and downloading course materials.
- Creating assignments, setting due dates, accepting submissions, and providing feedback or marks.
- Publishing and viewing announcements.
- Recording and viewing participation records according to permissions.
- Maintaining faculty activity entries related to course delivery.
- Dashboards, reminders, and basic academic activity reports.
- Audit information for important administrative changes.

3.2 Out of Scope

- Automated examination proctoring or biometric identification.
- Full learning management features such as live video conferencing or online examination engines unless added in a future version.
- Payroll, staff recruitment, fee collection, and institutional finance.
- Automatic grading of all assignment types.
- Integration with external student information systems unless explicitly approved.
- Guaranteeing that users read or understand every announcement.

3.3 Operating Environment

The proposed system is intended to operate as a web application accessible through a modern browser on desktop, laptop, tablet, or mobile devices. A server-side application and database store system records. Internet or institutional network access is required for normal remote use.

4. Functional Requirements

Each requirement is assigned a unique identifier to support implementation and testing. The system shall meet the following functional requirements.

ID	Feature	Requirement

FR-01	Authentication	The system shall allow registered users to log in using valid credentials.
FR-02	Access control	The system shall show functions and records according to the user's role and assigned permissions.
FR-03	Account management	The administrator shall be able to create, update, deactivate, and manage user accounts.
FR-04	Course management	The administrator shall be able to create and update courses, assign faculty, and manage student enrollment.
FR-05	Timetable management	Authorized faculty or administrators shall be able to create and update classroom schedules, including course, date, time, and room.
FR-06	Schedule viewing	Students and faculty shall be able to view schedules relevant to their courses and roles.
FR-07	Schedule change notification	The system shall notify affected users when an authorized schedule change is published.
FR-08	Material upload	Faculty shall be able to upload course materials with a title, description, course association, and file.
FR-09	Material access	Authorized students shall be able to view and download materials for courses in which they are enrolled.
FR-10	Assignment creation	Faculty shall be able to create assignments with instructions, release date, due date, allowed file types, and course association.
FR-11	Assignment submission	Students shall be able to submit assignments before the configured deadline and receive a submission confirmation.
FR-12	Submission status	The system shall display assignment status such as not submitted, submitted, late, or returned.
FR-13	Evaluation and feedback	Faculty shall be able to review submissions, record marks where applicable, and provide feedback.
FR-14	Announcement publishing	Faculty and authorized administrators shall be able to publish announcements for selected courses or user groups.
FR-15	Announcement viewing	Users shall be able to view announcements relevant to them, including publication time and author.
FR-16	Participation recording	Authorized faculty shall be able to record participation using institution-approved criteria.
FR-17	Participation review	Faculty shall be able to review participation records for their courses; students may view their own records if enabled.
FR-18	Faculty activity log	Faculty shall be able to record relevant academic activities such as material publication, assignment creation, and class-related updates.
FR-19	Dashboard	The system shall present role-specific summaries of upcoming classes, deadlines, announcements, and pending tasks.
FR-20	Reports	Authorized users shall be able to generate basic reports on schedules, assignment status, participation, and faculty activity.
FR-21	Search and filtering	The system shall allow users to search or filter relevant courses, materials, announcements, and assignments.
FR-22	Notifications	The system shall display in-app notifications for relevant new announcements, schedule changes, and assignment deadlines.
FR-23	Profile management	Users shall be able to view and update permitted profile details and change their password.
FR-24	Data backup and recovery	The system shall support routine data backup and restoration by authorized technical personnel.
FR-25	Audit trail	The system shall record important

		administrative actions, including account changes and course or schedule updates.

5. Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Performance	For normal usage, common pages should load within 3 seconds under the agreed test load; common save actions should return a response within 3 seconds, excluding slow network or file transfer conditions.
NFR-02	Security	The system shall require authentication for protected functions, enforce role-based access, store passwords using a secure one-way hashing method, and protect traffic with HTTPS in deployment.
NFR-03	Privacy	Users shall only access personal, course, submission, and participation data they are authorized to view.
NFR-04	Reliability	The system shall validate inputs and handle failures without corrupting previously saved records.
NFR-05	Availability	The target service availability should be at least 99% per month during agreed service hours, excluding planned maintenance.
NFR-06	Usability	The interface shall use clear labels, consistent navigation, readable text, and helpful validation messages.
NFR-07	Accessibility	Core workflows should support keyboard navigation, adequate contrast, and meaningful labels for form controls.
NFR-08	Scalability	The architecture should support adding courses and users without requiring a complete redesign.
NFR-09	Maintainability	The application shall use modular components, consistent naming, documentation, and version control to simplify updates.
NFR-10	Compatibility	The application should work with current versions of major browsers such as Chrome, Edge, Firefox, and Safari.
NFR-11	Data integrity	The system shall validate required fields, enforce valid relationships between users and courses, and prevent invalid due dates or duplicate enrollment records.
NFR-12	Backup and recovery	Backups shall be scheduled at an institution-approved interval, and restoration shall be tested periodically.
NFR-13	Auditability	Important changes shall record the acting user, action, and timestamp where appropriate.
NFR-14	Portability	The application should be deployable on a supported server environment with configuration separated from application code.
NFR-15	Legal and ethical use	The system shall follow applicable institutional policies and applicable privacy, accessibility, copyright, and data-protection requirements.

6. Actors and Use Cases

6.1 Primary Actors

Actor	Description	Major Use Cases
Administrator	Manages users, courses, enrollment, permissions, and system settings.	Log in; manage accounts; manage courses; assign faculty; manage enrollment; review reports; maintain settings.

Faculty Member	Manages teaching activities and course information.	Log in; view timetable; upload materials; create assignments; review submissions; publish announcements; record participation; record activity; generate course reports.
Student	Uses classroom information and learning resources.	Log in; view timetable; access materials; view announcements; submit assignments; view submission status and feedback; view permitted participation records.
Academic Coordinator	Reviews approved academic summaries and coordination information.	Log in; review schedules; view authorized reports; identify scheduling conflicts.

6.2 Secondary / Supporting Actors

- Notification service: supports delivery of in-app alerts and, if configured, email notifications.
- File storage service: stores uploaded learning materials and assignment submissions.
- Database service: stores user, course, schedule, assignment, participation, and activity records.
- Technical support / maintainer: performs backups, recovery, maintenance, and troubleshooting.

6.3 Actor–Use Case Summary

Actor	Use Cases
Administrator	Authenticate; manage users; manage courses; assign faculty; enroll students; configure permissions; review system reports; maintain settings.
Faculty Member	Authenticate; view timetable; upload materials; create assignment; review submission; publish announcement; record participation; maintain faculty activity; view reports.
Student	Authenticate; view timetable; access materials; view announcements; submit assignment; check submission status; view feedback; view permitted participation record.
Academic Coordinator	Authenticate; review timetable; inspect schedule conflicts; view authorized academic reports.
Notification service	Deliver configured alerts when relevant events occur.
File storage service	Store and retrieve permitted course files and assignment submissions.

6.4 Use Case Diagram

The diagram below shows the main actors and the major interactions with the Smart Classroom Management System. Supporting technical services are represented as external systems.

ACTORS	SMART CLASSROOM MANAGEMENT SYSTEM	SUPPORTING SERVICES
Administrator **Faculty Member** **Student** **Academic Coordinator**	**(Manage Users & Courses)** **(View / Manage Timetable)** **(Upload / Access Materials)** **(Create / Submit Assignments)** **(Review / Give Feedback)** **(Publish / View Announcements)** **(Record / Review Participation)** **(Manage Faculty Activities)** **(View Reports & Dashboard)**	**Notification Service** **File Storage Service** **Database Service**

Actor relationships: Administrator interacts with user/course management and reports; Faculty Member interacts with timetable, materials, assignments, announcements, participation, faculty activity, and reports; Student interacts with timetable, materials, announcements, assignments, feedback, and permitted participation records; Academic Coordinator interacts with schedules and authorized reports. Notification, file storage, and database services support the relevant use cases.

Note: This is a compact conceptual representation for the report. If a strict UML stick-figure diagram is required, the same actors and use cases can be redrawn using a UML tool such as diagrams.net or StarUML.

7. Use Case Descriptions

UC-01: Log In

Field	Description
Primary actor(s)	Student, Faculty Member, Administrator, Academic Coordinator
Precondition	The user needs access to the system.
Main flow	1. User opens the login page. 2. User enters credentials. 3. System validates credentials and role. 4. System opens the appropriate dashboard.
Alternative / exception flow	Invalid credentials display a safe error message; repeated failures may trigger rate limiting.
Related requirements	FR-01, FR-02

UC-02: Manage Timetable

Field	Description
Primary actor(s)	Administrator, Faculty Member; Student and Academic Coordinator view schedules
Precondition	An authorized user creates or updates a schedule.
Main flow	1. User opens timetable management. 2. User enters course, room, date, start time, and end time. 3. System validates required fields and conflicts. 4. System saves the schedule. 5. System makes it visible to affected users and issues a notification when appropriate.
Alternative / exception flow	Missing fields or overlapping room/time entries are flagged; unauthorized changes are rejected.
Related requirements	FR-05, FR-06, FR-07

UC-03: Upload Course Material

Field	Description
Primary actor(s)	Faculty Member
Precondition	Faculty wants to share learning material.
Main flow	1. Faculty selects a course. 2. Faculty enters a title and description. 3. Faculty selects a permitted file. 4. System checks file type and size. 5. System stores the file and lists it for enrolled students.
Alternative / exception flow	Unsupported or oversized files are rejected with a clear message.
Related requirements	FR-08, FR-09

UC-04: Submit Assignment

Field	Description
Primary actor(s)	Student; Faculty Member creates and reviews assignment
Precondition	A student submits work for an available assignment.
Main flow	1. Student opens the assignment. 2. Student reviews instructions and deadline. 3. Student uploads the required file or enters the permitted response. 4. System validates the submission. 5. System records timestamp and status. 6. System confirms successful submission.
Alternative / exception flow	If the deadline has passed, the system follows the configured late-submission policy and clearly identifies the submission as late or blocks it if late work is disabled.
Related requirements	FR-10, FR-11, FR-12

UC-05: Review Submission and Provide Feedback

Field	Description
Primary actor(s)	Faculty Member
Precondition	Faculty evaluates a student's submission.
Main flow	1. Faculty opens the assignment submissions. 2. Faculty selects a student submission. 3. Faculty reviews the work. 4. Faculty records marks and feedback if applicable. 5. System saves the evaluation and makes it visible to the student according to course settings.
Alternative / exception flow	Only authorized faculty for the course may evaluate submissions; invalid marks are rejected.
Related requirements	FR-13

UC-06: Publish Announcement

Field	Description
Primary actor(s)	Faculty Member, Administrator
Precondition	An authorized user shares a notice with a target audience.
Main flow	1. User selects the course or user group. 2. User enters announcement title and message. 3. User previews and publishes. 4. System records author and timestamp. 5. The announcement appears to the target audience and triggers configured notifications.
Alternative / exception flow	Empty messages or unauthorized target groups are rejected.
Related requirements	FR-14, FR-15, FR-22

UC-07: Record Participation

Field	Description
Primary actor(s)	Faculty Member; Student may view own record
Precondition	Faculty records participation for a class or activity.
Main flow	1. Faculty selects a course and class session. 2. Faculty selects students and records participation using approved criteria. 3. System validates the entries. 4. System saves records and displays a confirmation. 5. Authorized users can view relevant records.
Alternative / exception flow	Users cannot edit records outside their permissions; changes should be traceable.
Related requirements	FR-16, FR-17, FR-25

UC-08: Generate Report

Field	Description
Primary actor(s)	Administrator, Faculty Member, Academic Coordinator
Precondition	An authorized user needs an academic activity summary.
Main flow	1. User selects report type and date/course filters. 2. System verifies permissions. 3. System gathers relevant records. 4. System displays the report and provides an approved export option if enabled.
Alternative / exception flow	The system hides data outside the user's access scope and displays a message if no matching records exist.
Related requirements	FR-20, FR-21

8. Data Requirements

The system shall store the following major data entities. Data fields listed are indicative and may be refined during database design.

Entity	Indicative Data Fields
User	User ID, full name, institutional email, password hash, role, status, contact details where required, created/updated timestamps.
Course	Course ID, course code, course name, description, faculty ID,

	semester/term, status.
Enrollment	Enrollment ID, student ID, course ID, enrollment date, status.
Classroom Schedule	Schedule ID, course ID, faculty ID, room, date, start time, end time, recurrence if supported, status.
Course Material	Material ID, course ID, uploader ID, title, description, storage reference, file type, upload timestamp, visibility.
Assignment	Assignment ID, course ID, creator ID, title, instructions, release date, due date, allowed file types, maximum marks, late policy.
Submission	Submission ID, assignment ID, student ID, file/text reference, submitted timestamp, status, version number.
Evaluation / Feedback	Evaluation ID, submission ID, evaluator ID, marks, feedback, evaluation timestamp.
Announcement	Announcement ID, author ID, target course/group, title, message, publish timestamp, expiry if applicable.
Participation Record	Record ID, course ID, student ID, class session/date, participation status or score, recorder ID, timestamp, remarks if allowed.
Faculty Activity	Activity ID, faculty ID, course ID where relevant, activity type, description, activity date, created timestamp.
Notification	Notification ID, recipient ID/group, event type, message/reference, created timestamp, read status.
Audit Log	Log ID, actor ID, action, affected record type/ID, timestamp, outcome, limited technical context.

8.1 Data Validation and Retention

- Institutional email or another approved identifier should uniquely identify each user.
- Required fields must not be stored as blank values.
- A course enrollment must reference an existing student and course.
- Assignment due dates must follow the configured release and deadline rules.
- Uploaded files must comply with configured file type and size limits and be checked for security risks.
- Access to marks, submissions, participation, and personal data must be restricted by role.
- Retention and deletion periods should follow institutional policy and applicable law.

9. External Interface Requirements

9.1 User Interface

- Provide separate dashboards or navigation options for administrators, faculty, and students.
- Use consistent menus, headings, buttons, form labels, and confirmation messages.
- Display schedules in a readable calendar or list view.
- Organize materials and assignments by course.
- Show clear deadline, submission status, and feedback information.
- Provide validation messages close to the relevant input field.

9.2 Hardware Interface

No specialized hardware is required. Users need a supported computer, tablet, or smartphone with a display, input method, and network connectivity. Server hardware or a cloud hosting environment must meet the selected deployment requirements.

9.3 Software Interface

- A modern web browser is used to access the system.
- A server-side application communicates with a relational database or another approved persistent data store.

- A file storage component stores course materials and assignment submissions.
- An optional email or push-notification provider may deliver alerts if configured by the institution.
- Any external interface must use authenticated, documented communication and handle service failures safely.

9.4 Communication Interface

The deployed application should use HTTPS for browser-to-server communication. Requests and responses should use standard web protocols and validated data formats. File uploads should be subject to size limits, type validation, and access checks.

10. System Features

Feature	Description
Feature 1: Role-Based Dashboard	Presents relevant schedule, deadlines, announcements, and pending actions based on the signed-in user's role.
Feature 2: Timetable Coordination	Allows authorized schedule management and provides current schedule information to affected users.
Feature 3: Course Material Repository	Stores course resources in an organized structure and provides controlled access to enrolled students.
Feature 4: Assignment Lifecycle	Supports assignment creation, student submission, status tracking, evaluation, and feedback.
Feature 5: Announcement Centre	Publishes course-level or group-level announcements and supports configured notifications.
Feature 6: Participation Tracking	Records and reviews participation according to approved academic practices and role permissions.
Feature 7: Faculty Activity Management	Maintains relevant academic activity entries and supports summary views.
Feature 8: Reporting and Search	Helps authorized users locate information and review basic academic and operational summaries.
Feature 9: Security and Audit	Protects restricted data and records important actions for accountability.

11. Assumptions and Constraints

11.1 Assumptions

- The institution will provide accurate course, student, faculty, and timetable information.
- Each user will have a unique account and will keep login credentials confidential.
- Faculty members and administrators will receive appropriate training or basic usage guidance.
- Users will have access to a compatible browser and a usable network connection.
- The institution will define assignment late policies, participation criteria, data retention rules, and notification preferences.
- Only authorized staff may approve or modify official academic records.
- External email or notification delivery depends on the availability and configuration of the chosen provider.

11.2 Constraints

Constraint Type	Constraint / Impact
Technical	The system depends on supported browsers, server capacity, database availability, file storage, and network connectivity.
Operational	User roles, course assignments, deadlines, and participation policies must be defined by the institution.

Economic	Hosting, domain, storage, backups, and optional notification services must fit the available budget.
Legal / ethical	Personal data, student submissions, marks, and participation records must be handled under applicable law and institutional policies; copyright rules apply to shared materials.
Security	Access controls, secure password handling, HTTPS, backups, and safe file upload practices are required.
Time-related	The project must be designed, implemented, tested, documented, and submitted within the academic schedule.
Scope	The first version focuses on classroom coordination and does not include full video conferencing, online examination, or automated proctoring.
Data quality	Reports are only as accurate as the schedule, enrollment, submission, and participation data entered into the system.

12. Requirement Validation

12.1 Completeness Check

Validation Question	Result / Evidence
Are all major stakeholders represented?	Yes. Administrator, faculty, students, academic coordinator, and supporting services are identified.
Are the main classroom activities covered?	Yes. Timetables, materials, assignments, announcements, participation, faculty activity, notifications, and reports are included.
Are both functional and quality requirements specified?	Yes. Functional requirements and measurable or testable non-functional requirements are listed separately.
Are use cases linked to requirements?	Yes. Each detailed use case includes related requirement identifiers.
Are data and interfaces described?	Yes. Core entities and user, software, hardware, and communication interfaces are described.
Are assumptions and constraints documented?	Yes. Technical, operational, economic, legal/ethical, security, scope, and schedule constraints are listed.

12.2 Consistency and Ambiguity Review

- Role consistency: permissions are consistently based on administrator, faculty, student, and coordinator roles.
- Terminology: 'course material', 'assignment', 'announcement', 'participation record', and 'faculty activity' are used consistently.
- Deadline handling: late submission behavior is controlled by an explicit course or assignment policy rather than an unstated universal rule.
- Privacy: student marks, submissions, and participation records are only visible to authorized users.
- Notifications: delivery beyond in-app alerts depends on configuration; the system must not promise external delivery when the provider is unavailable.
- Performance targets: response-time and availability goals are intended to be verified under an agreed test environment and workload.

12.3 Potential Gaps and Improvements

Potential Gap	Recommended Improvement
Detailed grading rules may vary by course.	Define whether marks, rubrics, resubmissions, and grade release are required for the first release.
Notification delivery channels may differ.	Confirm whether email, push notifications, or in-app notifications are mandatory.
Participation criteria can be subjective.	Agree on a transparent set of participation categories and who may edit records.
Schedule conflicts need policy decisions.	Specify whether room conflicts, faculty conflicts, and student timetable conflicts must be detected automatically.
File size and allowed formats are not yet fixed.	Set institution-approved upload size limits, formats, and malware scanning rules.

Retention periods depend on policy.	Document retention, archiving, export, and deletion periods for submissions and academic records.
Availability target requires operational agreement.	Confirm hosting, maintenance windows, backup frequency, recovery time, and recovery point objectives.

12.4 Acceptance Criteria

- A user can log in and only access features allowed for their role.
- An administrator can create a course and assign faculty and students.
- Authorized users can create and view a timetable, and relevant users can see schedule changes.
- Faculty can upload materials and students enrolled in the course can access them.
- Faculty can create an assignment and a student can submit it and receive a confirmation.
- Faculty can review a submission and provide feedback that is visible to the intended student.
- Authorized users can publish announcements and target users can view them.
- Faculty can record participation and unauthorized users cannot alter those records.
- Authorized users can generate reports without seeing data outside their permissions.
- Validation errors, unavailable services, and unauthorized requests are handled clearly and safely.

13. Conclusion

The Smart Classroom Management System is proposed as a centralized platform for classroom schedules, course materials, assignments, announcements, student participation, and faculty activities. This SRS identifies the stakeholders, system scope, functional and non-functional requirements, actors, use cases, data entities, interfaces, assumptions, constraints, and validation criteria. The specification provides a structured foundation for design, implementation, testing, and future enhancement. Before development begins, the institution should confirm configurable policies such as late submissions, participation categories, file limits, notification channels, and data retention.

Appendix A: Requirement Traceability Snapshot

Requirement Area	Related Use Cases / Features	Validation Evidence
Authentication and permissions	UC-01; Feature 1; Feature 9	Role-based access tests and invalid-login tests.
Timetable	UC-02; Feature 2	Create/view/update tests and conflict validation.
Materials	UC-03; Feature 3	Upload/download tests and access-control checks.
Assignments	UC-04, UC-05; Feature 4	Submission, deadline, feedback, and status tests.
Announcements	UC-06; Feature 5	Audience targeting and notification tests.
Participation	UC-07; Feature 6	Create/view/edit permission tests.
Reports	UC-08; Feature 8	Filter accuracy and authorization tests.
Data integrity and audit	Features 7 and 9	Required-field, relationship, and audit-log tests.